
Mobile and Ubiquitous Computing

Motivation, Scenarios, Problems, Concepts



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Why do I need to learn MUC?

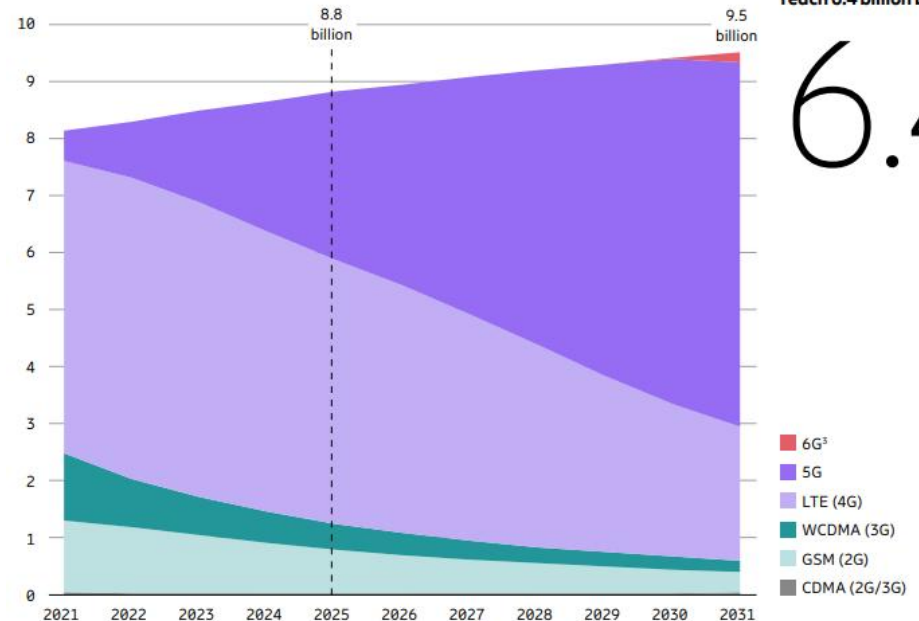
Why do I need to learn smartphone programming?

According to the [Ericsson Mobility Report](#) (June 2026)

There are nearly 8.7 billion mobile subscriptions worldwide

About 90% (7.8bn) are mobile broadband subscriptions. Under 4.6 billion are 4G subscriptions, while 5G subscriptions have reached nearly 3.3 billion

Figure 1: Mobile subscriptions by technology (billion)



5G subscriptions are forecast to reach 6.4 billion by the end of 2031.

6.4bn

This is equivalent to about 106% of the world population (8.2bn)

Why do I need to learn MUC?

Why do I need to learn smartphone programming?

What will drive our research in mobile computing over the next decade?

One force is the ***emergence of mobile devices as rich sensors***

The other force is the ***convergence of mobile computing and cloud computing***

-- Mahadev Satyanarayanan

Motivation

- **Need for access to information**
 - Autonomy in the access
 - Temporal and geographic independence
- **Increased work efficiency**
 - We communicate while we work
 - Connectivity anywhere, anytime, while we walk, run
- **Lower installation cost & infrastructure maintenance**
 - Antenna placement vs cable
 - Dynamic allocation of communication channels
- **Mobile devices (smartphones, tablets, smart *, ...)**
 - Reduction of cost, size, and weight
 - More features and functionalities
 - Portability



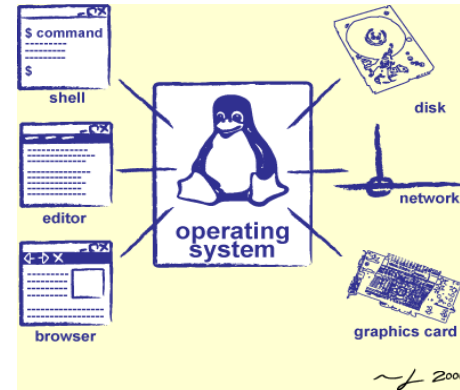
Motivation

- Most devices are becoming **wireless**
 - e.g., phones, computers, cameras, sensors, home appliances,...
- Most services are becoming **mobile**
 - e.g., e-mail, web, TV,...
- People are more and more mobile
 - need for **nomadic** technologies
- Networks are growing to accommodate the needs of a mobile society
 - larger numbers of users, larger bandwidth, larger coverage, interoperability,...
- New devices, applications, and services are emerging

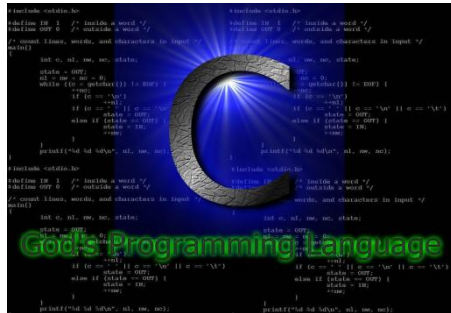
What does this encapsulate?



User-interface design



Operating system design



programming languages



networking



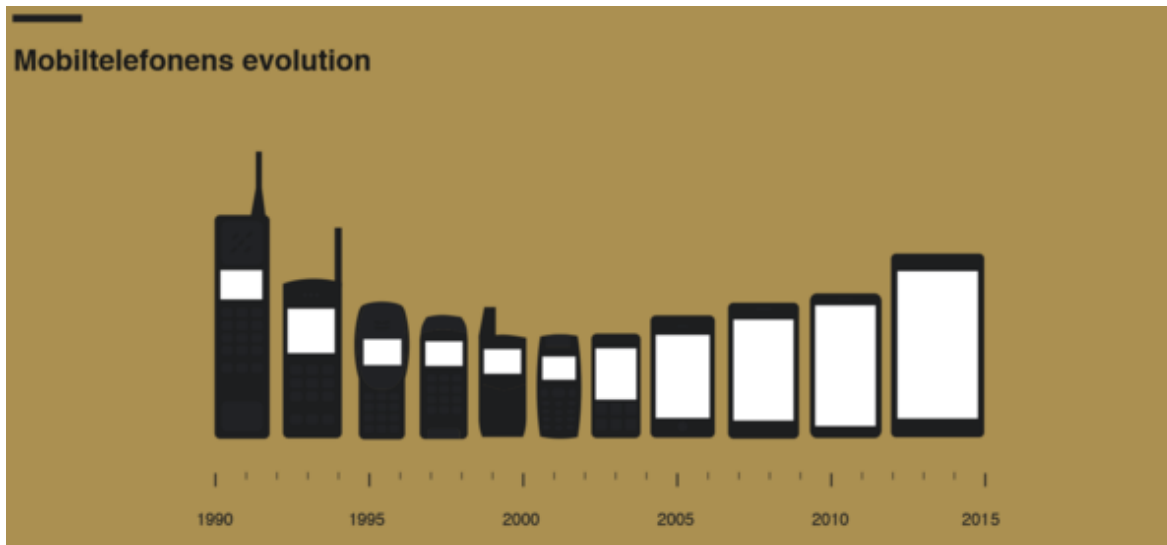
cloud computing

Why is it difficult to develop good Mobile Apps?

- Computational power
 - From quad-core@ 1.8 GHz, to octa-core/deca-core@ 3-4.7 GHz
 - What if I need more processing power: use the cloud?
 - Still relevant. Dedicated cores NPU (AI)
- Memory (can be limited for certain applications)
 - 4 GB to 12-16 GB of RAM (flagship)
- Battery constraint (still the **biggest constraint**)
 - Limited battery is always a constraint
 - Cannot build apps that kill your battery
- Too many options 😊
 - Face ID, Accelerometer, GPS, Compass, Gyroscope, Proximity sensor, Ambient light sensor
 - WiFi, Bluetooth, ... 5G.... early 6G research (not yet deployed)

The Screens evolved ... but

- From phone to smartphones
- Can we go so small/so big?
- New devices are emerging



The specification of a smartphone!



Sensors

Face ID
Barometer
High dynamic range gyro
High-g accelerometer
Proximity sensor
Dual ambient light sensors

Power and Battery³

Video playback
Up to 30 hours

Video playback (streamed)
Up to 27 hours

Built-in rechargeable lithium-ion battery

Fast-charge capable:

Up to 50% charge in 20 minutes⁴ with [40W adapter](#) or higher (available separately) paired with USB-C charging cable

Up to 50% charge in 30 minutes⁴ with 30W adapter or higher paired with MagSafe Charger (both available separately)

Cellular and Wireless

Model A3258^{5,6}

FDD-5G NR (Bands n1, n2, n3, n5, n7, n8, n12, n14, n20, n25, n26, n28, n29, n30, n66, n70, n71, n75)

TDD-5G NR (Bands n38, n40, n41, n48, n53, n77, n78, n79)

5G NR mmWave (Bands n258, n260, n261)

FDD-LTE (Bands 1, 2, 3, 4, 5, 7, 8, 12, 13, 14, 17, 18, 19, 20, 25, 26, 28, 29, 30, 32, 66, 71)

TDD-LTE (Bands 34, 38, 39, 40, 41, 42, 48, 53)

UMTS/HSPA+/DC-HSDPA (850, 900, 1700/2100, 1900, 2100 MHz)

GSM/EDGE (850, 900, 1800, 1900 MHz)

5G (sub-6 GHz and mmWave) with 4x4 MIMO⁸

Gigabit LTE with 4x4 MIMO⁹

Apple N1 wireless networking chip

Wi-Fi 7 (802.11be) with 2x2 MIMO¹⁰

Bluetooth 6

Thread networking technology

Apple second-generation Ultra Wideband chip¹¹

NFC with reader mode

Express Cards with power reserve

Location

Precision dual-frequency GPS (GPS, GLONASS, Galileo, QZSS, BeiDou, and NavIC)

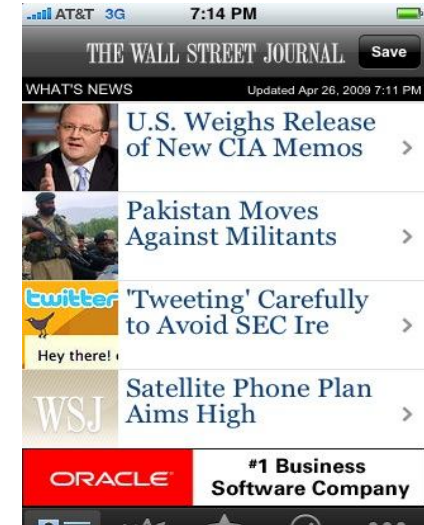
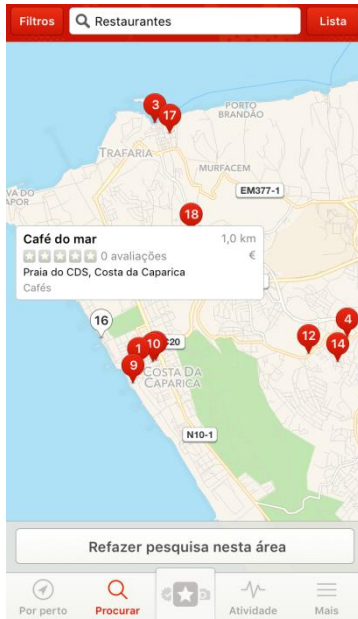
Digital compass

Wi-Fi

Cellular

iBeacon microlocation

What are the potential applications/services endless...



Scenarios some

- Creation of communication groups for
 - Problem solving
 - Emergency services
 - Rescue services
 - Military operations
 - Decision support systems
 - Generation of ideas



Scenarios emergencies

Ad-Hoc networks
Heterogeneity

- early transmission of patient data to the hospital, current status, first diagnosis
- replacement of a fixed infrastructure in case of earthquakes, hurricanes, fire, ...
- crisis, war, ...



Scenarios ... Field work

Commercial services, Entertainment, Health, Education

- Data access from mobile devices
- Easy retrieval and collection of data
- Direct access to customer/patient files stored in a central location
- Consistent databases for all agents/users
- Mobile office



Data replication
(no | limited) **Connectivity**
Heterogeneity
Adaptation

More Scenarios ... entertainment

- Creation of communication groups for
 - Entertainment
 - multiuser games,
 - data exchange (music, movies, ...)

Ad-Hoc networks
Heterogeneity



Mobile/Nomadic computing

- Nomadic computing
 - is a technology allowing anyone to leave their office and still have seamless access to the same set of network services as they had at their office, wherever they go with whatever device they're carrying, regardless of the environment they enter.” -- Leonard Kleinrock
- Mobile/Nomadic computing (1980)
 - Paradigm in which users could carry their personal computers and retain some connectivity to other machines

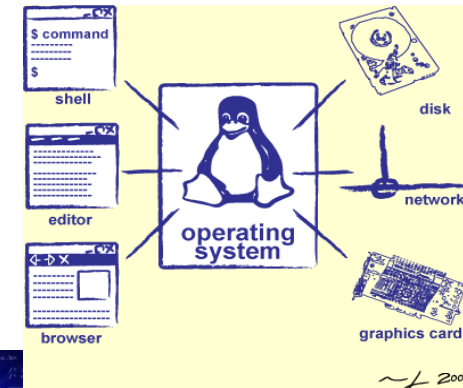
Mobile computing

- Use of
 - **Mobile** computing devices - systems are *portable and can be used by people on the move*
- For
 - The execution of stand-alone or connected applications through wireless networks
 - **Wireless**: communication with other systems *does not require wires*
- By
 - Mobile users, with different paces, coverage... context
 - (also) No users...

Wireless communication, mobility, heterogeneity, data processing

Mobile computing perspectives

- The technology perspective
 - The hardware
- The system perspective
 - Networks, Operating Systems
- The application
 - Data and representation
- The user
 - Interaction
- The programming perspective



Ubiquitous computing

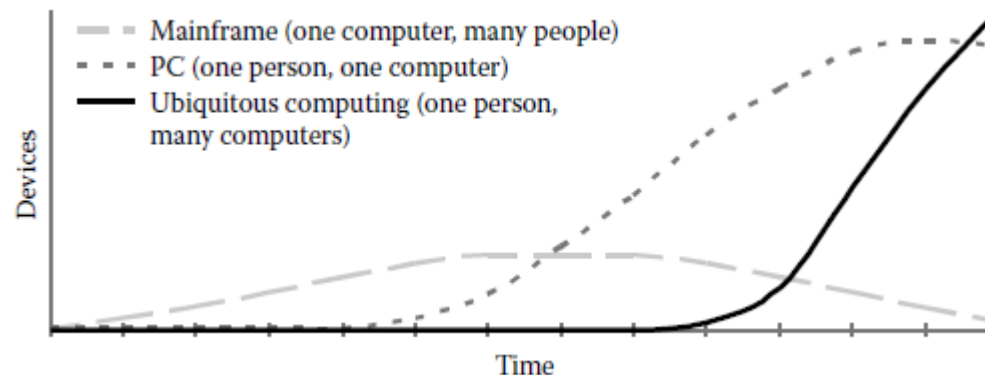
"Ubiquitous computing names the third wave in computing, just now beginning.

First were mainframes, each shared by lots of people.

Now we are in the personal computing era, person and machine staring uneasily at each other across the desktop.

Next comes **ubiquitous computing**, or the age of calm technology, when technology recedes into the background of our lives."

Mark Weiser, 1989



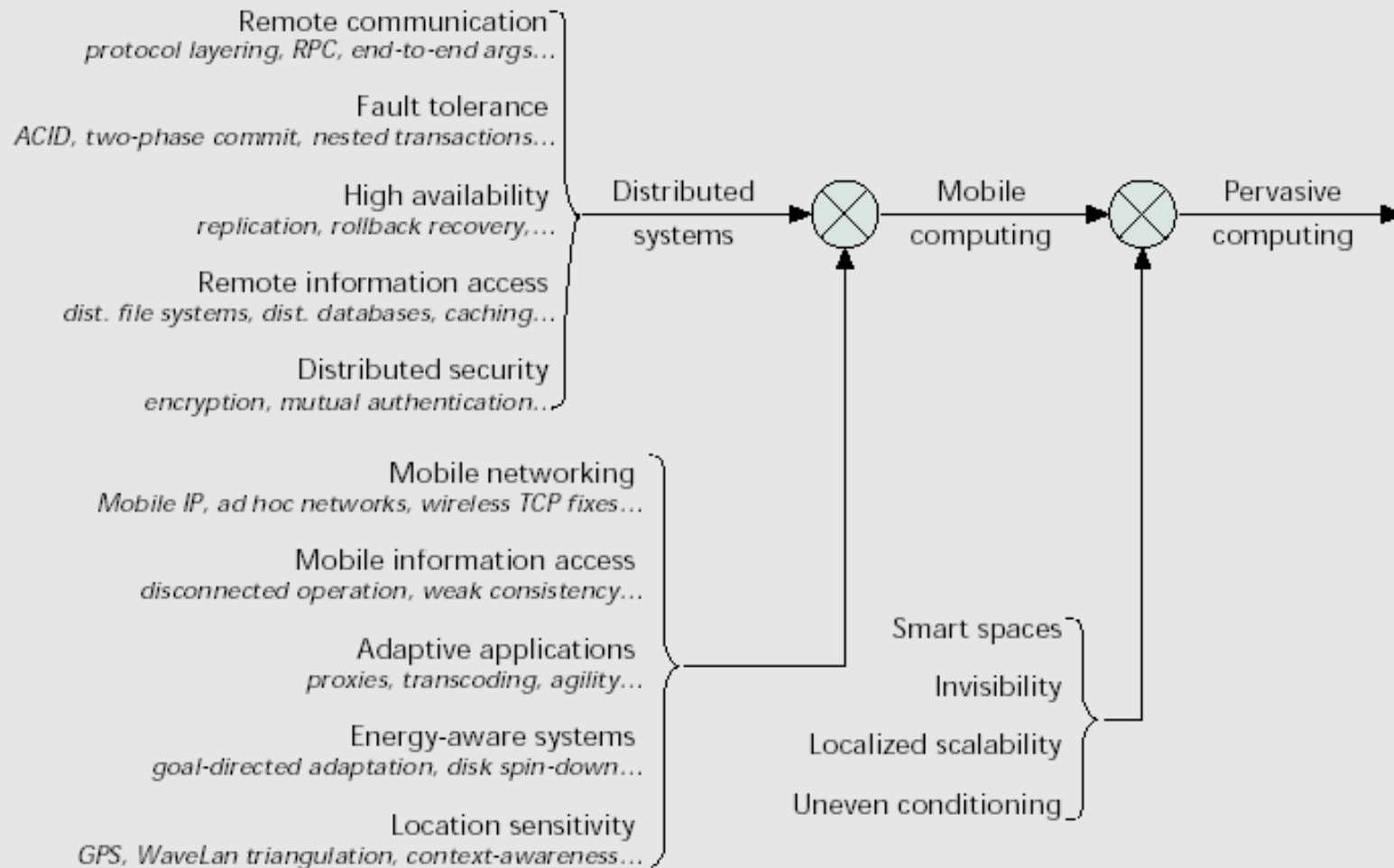
Ubiquitous computing

- An interesting definition (Princeton)
 - is a post-desktop model of human computer interaction in which **information processing** has been thoroughly **integrated** into **everyday objects** and **activities**
 - machines that fit the human environment instead of forcing humans to enter theirs
 - Integrate computers seamlessly into the world
invisible, everywhere computing
often called pervasive/invisible computing

Ubiquitous computing

- Technology View
 - Computers everywhere – embedded into fridges, washing machines, door locks, cars, furniture
 - Intelligent environment
 - Mobile portable computing devices
 - Wireless communication – seamless mobile/fixed
- User View
 - Invisible – implicit interaction with your environment
 - Augmenting human abilities in context of tasks

Relation between DS, MC and PC



Fonte: Satyanarayanan, Pervasive Computing: Vision and Challenges, IEEE Personal Communications, 2001

Pervasive computing

- **Smart spaces** - A simple example of this is the automatic adjustment of heating, cooling, and lighting levels in a room based on an occupant's electronic profile
- **Invisibility** - complete disappearance of pervasive computing technology from a user's consciousness
- **Localized scalability** - As smart spaces grow in sophistication, the intensity of interactions between a user's personal computing space and his/her surroundings increases. This has severe bandwidth, energy, and distraction implications for a wireless mobile user
- **Masking Uneven Conditioning** — The fourth thrust is the development of techniques for masking uneven conditioning of environments. The rate of penetration of pervasive computing technology into the infrastructure will vary considerably depending on many nontechnical factors such as organizational structure, economics, and business models

More on related definitions

- Wearable computing
 - Devices attached to clothes, worn like watches, jewelry, ...
 - Now mainstream: Apple Watch, Meta Ray-Ban Display, Apple Vision Pro, Oura Ring, WHOOP
- Context-aware computing
 - e.g: device will automatically switch itself to “vibrate” instead of “ring” when it is in the cinema



Context, context and context

